Executive Summary

- **Five concise PDFs** will cover (1) Global Superstore dataset, (2) key retail KPIs, (3) business use cases, (4) sample insights/dashboards, and (5) sales strategies/actions.
- **Section 1 (Dataset):** describes tables (Orders, Returns, People) including columns, types, sample, relationships and cleaning.
- **Section 2 (KPIs):** defines major metrics (Revenue, Gross Profit, Profit Margin, AOV, CLV, Order Freq., Return Rate, Fill Rate, Inventory Turnover) with formulas, interpretation and example SQL/Excel formulas.
- **Section 3 (Use Cases):** lists top retail problems (demand forecast, inventory issues, pricing, segmentation, returns, fulfillment, etc.), with hypotheses, data signals and analysis methods[1][2].
- **Section 4 (Insights/Dashboards):** gives 10 example insights (e.g. sales trends, category profits, churn by segment), chart recommendations and a mock dashboard flow (mermaid chart).
- **Section 5 (Strategies):** offers actionable strategies mapped to data signals (e.g. increase AOV, reduce returns), plus A/B test ideas and a 2×2 prioritization matrix of impact vs. effort.

Each PDF section will include 2–3 royalty-free images (with search keywords, source, caption, placement). We list suggested PDF filenames, estimated pages, and chunking strategy for RAG ingestion (size, overlap, metadata). All facts are cited from authoritative sources.

Dataset Explanation (PDF 1)

Figure: Printout of spreadsheet numbers – illustrates raw data tables. *(Image: search "spreadsheet numbers data unsplash")*

- **Tables & Size:** The Global Superstore data is an Excel workbook (2011–2014) with three sheets: *Orders*, *Returns*, and *People*[2]. Orders has ~51,291 rows and 24 columns[2]; Returns has 1,174 rows (3 cols); People has 14 rows (2 cols).
- **Columns & Types:** Key Order fields include Order ID, Order Date, Ship Date, Ship Mode, Customer ID/Name, Segment, City/State/Country, Market, Region, Product ID/Category/Sub-Category/Product Name, Sales, Quantity, Discount, Profit, Shipping Cost, and Order Priority[3][4]. Data types mix *string* (IDs, names, categories), *datetime* (dates), and numeric (Sales, Profit, Quantity, Discount).
- **Relationships:** Each Orders record links to *Returns* via Order ID (left join; Orders as primary)[5]. The *People* sheet holds only manager info – typically not needed for analysis.
- **Sample Row:** (e.g.) an order row: `OrderID=CA-2016-152156, OrderDate=11/8/2016, CustomerName=Claire Gute, Segment=Consumer, City=Henderson, State=Kentucky, Country=USA, Region=East, ProductID=FUR-BO-10001798, Category=Furniture, Sub-Category=Bookcases, ProductName=Bush Somerset Collection Bookcase, Sales=261.96, Quantity=2, Discount=0, Profit=41.9136, ShippingCost=10.60, OrderPriority=Medium.` (formatted table in PDF).
- **Data Quality:** Fields are nearly complete – e.g. only PostalCode ~0.8% null[4]; no duplicate Order IDs[4]. Suggest drop unused columns (Row ID, Order ID, Customer ID, Product ID, Postal Code) and compute any needed fields (e.g. *Profit Ratio*=Profit/Sales). Check date consistency (Order Date ≤ Ship Date).
- **Preprocessing:** Format date fields, trim whitespace in text, encode categories (e.g.

numeric codes), and correct data types. Normalize product categories and region names. Handle any duplicates (remove or aggregate by highest priority). Consider partitioning by year for efficient querying.

Column	**Type**	**Sample Values**
Order ID	string	CA-2016-152156
Order Date	date	2016-11-08
Ship Mode	string	Second Class
Customer Name	string	Claire Gute
Segment	string	Consumer
City	string	Henderson
State	string	Kentucky
Country	string	United States
Region	string	East
Market	string	US
Product Category	string	Furniture
Sub-Category	string	Bookcases
Product Name	string	*Bush Somerset Collection Bookcase*
Sales	float	261.96
Quantity	int	2
Discount	float	0.00
Profit	float	41.91
Shipping Cost	float	10.60
Order Priority	string	Medium

- **Image Suggestions:**
- *Spreadsheet printout:* (“spreadsheet print data”, **Unsplash**, embed top).
- *Computer with code:* (“computer code screen”, **Unsplash**, embed near text on ETL).
- *Excel on laptop:* (“laptop excel data Pexels”, **Pexels**).

Filename: `global_superstore_dataset.pdf` (≈4 pages).
RAG Chunking: 500 tokens per chunk, 50-token overlap; metadata: {`source`:“dataset_expl”, `page`, `section`}.

KPI Definitions & Calculation Guide (PDF 2)

Figure: Computer screen with code – e.g. KPI formulas in SQL/Excel. *(Image: “computer code unsplash”)*

- **Revenue:** Total sales. *Formula:* `SUM(Sales)`. *SQL/Excel:* e.g. `SELECT SUM(Sales) FROM Orders;` or `=SUM(Table[Sales])`. *Interpretation:* Gross income before costs. Aim: track growth over time or vs. targets.
- **Gross Profit:** Revenue minus COGS (if COGS unknown, use Profit field or omit COGS). *Formula:* `Sales - COGS`. *Example:* `GrossProfit = Sales - COGS`. *Interpretation:* Money left after covering production costs[6].
- **Profit Margin (Net):** Profit as % of revenue. *Formula:* `(Profit ÷ Sales)×100`. *SQL:* `100.0 * SUM(Profit) / SUM(Sales)`. *Excel:* `=SUM(Profit)/SUM(Sales)`. *Interpretation:* Efficiency measure – how much of sales is profit. (Goal: higher is better; retail often low, e.g. 2–10%[6][7].)
- **Average Order Value (AOV):** *Formula:* `Total Revenue ÷ Number of Orders`[8]. *SQL:* `AVG(order_amount)` or `SUM(Sales)/COUNT(DISTINCT OrderID)`. *Excel:* `=SUM(Sales)/COUNTA(OrderID)`. *Interpretation:* Customer spending per order. Higher AOV implies stronger sales per transaction[8].
- **Customer Lifetime Value (CLV):** *Formula:* `Avg Order Value × Orders per Year × Customer Lifespan`. E.g. CLV =$50×26×7 = $9,100[9]. *Interpretation:* Total revenue expected from a customer. Helps segment high-value customers[9].
- **Order Frequency:** *Formula:* `Total Orders ÷ Number of Unique Customers (per period)`. Or measure avg. orders/year/customer. *SQL:* `COUNT(orderID)/COUNT(DISTINCT CustomerID)`. *Interpretation:* Buying rate per customer. (Increasing freq. boosts CLV[10].)

- **Return Rate:** *Formula:* `(Number of Returned Orders ÷ Total Orders)×100`. *Example:* If 200 orders returned of 10,000, return rate=2%. *Interpretation:* % of sales undone by returns. Lower is better; target ≈ <5%.
- **Fill Rate:** *Formula:* `(Orders filled completely ÷ Total orders)×100`. *Interpretation:* Service level – % orders shipped in full on first attempt. Best-in-class ≈95–98%[11]. *SQL/Excel:* e.g. `100 * SUM(CASE WHEN FillStatus='Full' THEN 1 ELSE 0 END)/COUNT(*)`.
- **Inventory Turnover:** *Formula:* `Cost of Goods Sold / Avg. Inventory`[12]. (Since COGS may be unknown, use `Sales / Avg. Inventory`). *Interpretation:* How many times inventory cycles per period. Higher is good (fast selling). *Excel:* `=COGS/AVERAGE([Inventory levels])`.

KPI	Formula (SQL/Excel)	Interpretation / Threshold
Revenue	`SUM(Sales)` / `=SUM(Sales)`	Total sales. Baseline metric for size of business.
Gross Profit	`SUM(Sales-COGS)` / `=SUM(Sales)-SUM(COGS)`	Sales minus cost of goods[6]; higher is better.
Profit Margin	`100*SUM(Profit)/SUM(Sales)` / `=SUM(Profit)/SUM(Sales)`	% profit of sales[7]; target depends on segment (often ~5–20%).
AOV	`SUM(Sales)/COUNT(DISTINCT OrderID)` / `=SUM(Sales)/COUNT(Orders)`	Avg. spend per order[8]; higher suggests high-value orders.
CLV	`(AOV * Orders/yr * lifespan)` / (custom calc)	Total revenue per customer[9]; use to identify valuable segments.
Order Frequency	`COUNT(Orders)/COUNT(DISTINCT CustomerID)`	Orders per customer (yr)[10]; increasing is key for loyalty.
Return Rate	`100*SUM(Returns)/SUM(Orders)` / `=COUNT(Ret)/COUNT(Orders)`	% orders returned; lower is better (target <5%).
Fill Rate	`100*SUM(Filled)/COUNT(Orders)`	% orders fulfilled in full[11]; aim ≥95% for service.
Inv. Turnover	`COGS / ((BegInv+EndInv)/2)` / `=COGS/AVERAGE(InvRange)`	How fast stock sells[12]; higher implies efficient inventory use.

- **Image Suggestions:**
- *Data charts:* (“financial charts pen”, **Pexels**, embed near KPI list).
- *KPI dashboard:* (“dashboard charts laptop”, **Unsplash**).
- *Calculator and report:* (“calculator ledger Pexels”, **Pexels**).

Filename: `global_superstore_kpis.pdf` (≈3 pages).
RAG Chunking: 450 tokens, 50 overlap; metadata: {`source`:"kpis", `page`, `section`}.

Business Problems & Use Cases (PDF 3)

Figure: Warehouse shelves – highlights inventory/stock challenges. *(Image: "warehouse shelves Pexels")*

Top Retail Problems for Global Superstore:

1. **Demand Forecasting & Stockouts:** Hypothesis: poor forecasting causes stockouts or overstock. *Signals:* low fill rates, frequent out-of-stock flags, high emergency restock costs. *Analysis:* Time-series sales by SKU, seasonality trends. (Predictive analytics can cut stockouts ~20–35%[1].)
2. **Excess Inventory / Turnover:** Hypothesis: slow-moving stock ties capital. *Signals:* low inventory turnover ratio, high warehouse values. *Analysis:* Age-stock report; promote or bundle slow SKUs.
3. **Pricing & Promotion Impact:** Hypothesis: suboptimal pricing erodes margins. *Signals:* high discount vs. low profit margins, margin variance by product. *Analysis:* Price elasticity models, margin lift tests. (Optimized pricing can boost margins ~3–8%[13].)
4. **Category Profitability:** Hypothesis: some categories/regions underperform. *Signals:* uneven profit per category, high returns in category, or negative profit products. *Analysis:* Pareto charts of sales vs profit by category/sub-category.
5. **Customer Segmentation & Retention:** Hypothesis: a few customers generate most profit. *Signals:* skewed sales/CLV distribution; low repeat purchase rate. *Analysis:* RFM analysis, cluster by order frequency/value, targeted retention strategies. (High-CLV

segments identified via CLV metric[9].)
6. **High Return Rates:** Hypothesis: some products or segments incur many returns, hurting margin. *Signals:* return rate spikes by category or product. *Analysis:* Drill into return reasons, correlate returns with discounts or promos.
7. **Fulfillment & Delivery Delays:** Hypothesis: shipping delays reduce satisfaction. *Signals:* low OTIF (on-time in-full) and fill rate metrics[11]. *Analysis:* Drill delay by region or carrier; improve logistics or safety stock.
8. **Seasonal / Channel Variability:** Hypothesis: sales vary strongly by season or channel (online vs store). *Signals:* large swings in month/quarter sales, inventory buildup pre-season. *Analysis:* Year-over-year trend analysis, pivot by channel.

For each problem, define a hypothesis (what we expect), identify data signals/KPIs to monitor, and plan analysis (e.g. trend charts, regression, clustering). These use cases align with industry best practices in retail analytics[1][2].

- **Image Suggestions:**
- *Warehouse inventory:* (“warehouse boxes Pexels”, **Pexels**, embed).
- *Retail shelf/display:* (“store shelves unsplash”, **Unsplash**).
- *Team meeting:* (“business discussion Pexels”, **Pexels**).

Filename: `global_superstore_problems.pdf` (≈4 pages).
RAG Chunking: 500 tokens, 50 overlap; metadata: {`source`:"use_cases", `page`, `section`}.

Sample Insights & Dashboards (PDF 4)

Figure: Team meeting with laptops – illustrates dashboard review. *(Image: “team meeting Pexels”)*

- **Example Insights:**

1. *Sales Trend by Year:* line chart of monthly/yearly sales to spot growth or seasonality.
2. *Profit by Category:* bar chart showing which product categories yield highest/lowest profit.
3. *Top Customers:* Pareto chart of customers by revenue or profit, highlighting VIPs.
4. *CLV Segments:* histogram of customer lifetime values; identify tail of high-value shoppers.
5. *Order Frequency Distribution:* bar chart of customer purchase frequency (1×, 2–5×, 6+× per year).
6. *Discount vs Profit:* scatter plot or bar showing how heavy discounts impact margins by product.
7. *Return Rate by Product:* heatmap of return rate % across sub-categories.
8. *Fill/OTIF by Region:* gauge or colored map showing regions meeting order fulfillment targets.
9. *Inventory Aging:* stacked bar of inventory age buckets (0–30d, 31–60d, etc.) for key products.
10. *Sales vs Forecast:* dual-axis line chart comparing actual vs predicted sales.

- **Chart Types:** Use a mix of time-series lines, bar charts, Pareto charts, heatmaps, and maps. Include filters (e.g. by region, category) to drill down.

- **Dashboard Layout:** Mock-up with KPI scorecards on top (Revenue, AOV, Fill%), key trend charts in center (Sales & Profit over time), and drill-down visuals on bottom (Category performance, Customer segments).

```
flowchart LR
    A[Raw Data: Orders/Returns] --> B[ETL/Modeling] --> C[Data Warehouse]
    C --> D[Sales KPI Dashboard]
    C --> E[Inventory & Order Dashboard]
    C --> F[Customer & Marketing Dashboard]
```

Figure: Data flow for dashboards. Raw data is cleaned/processed (ETL), loaded into a warehouse, and queried by dashboards. (Mermaid chart above).

- **Image Suggestions:**
- *Charts on desk:* ("charts laptop magnifying glass Pexels", **Pexels**, embed).
- *Dashboard screenshot:* ("business dashboard unsplash", **Unsplash**).
- *Flow diagram:* ("data flowchart whiteboard unsplash", **Unsplash**).

Filename: `global_superstore_insights.pdf` (≈4 pages).
RAG Chunking: 450 tokens, 50 overlap; metadata: {`source`:"insights", `page`, `section`}.

Sales Strategies & Prescriptive Actions (PDF 5)

Figure: Office meeting – strategizing actions. *(Image: "business meeting Pexels")*
- **Actions Mapped to Signals:**
- *Low AOV:* Introduce bundles or upsell prompts. **Test:** A/B test product bundling vs. single

sales.
- *High Return Rate:* Improve product descriptions or quality. **Action:** Enhance QC or liberal exchange policy for problem items. **Test:** Offer free returns coupon to some customers vs. control.
- *High Discount/Low Margin:* Review pricing; launch loyalty to shift discounts to loyalty points. **Test:** 20% discount coupon vs. loyalty points equivalent.
- *Under-stock in Fast Movers:* Increase safety stock and improve forecasting. **Test:** Priority reorder rule vs. fixed schedule.
- *Slow-moving Inventory:* Run clearance promotions or reallocate stock. **Action:** Flash sale on old SKUs; measure sell-through.
- *Low CLV Segment:* Introduce targeted marketing or subscription offers to boost repeat buys. **Test:** Personal email campaign vs. generic promo.
- *Underperforming Regions:* Local promotions or adjust assortment to local tastes. **Action:** Tailored bundles per region.
- *Channel Shift:* If e-commerce lags brick-and-mortar, invest in online UX or exclusive web-only deals.

- **Prioritization:** Plot actions on a 2×2 value/effort matrix (Value=impact on KPIs; Effort=cost/time). For example, a "reach new customers" loyalty program might be high-impact but high-effort, whereas "improve product descriptions" is low-effort medium-impact. Focus first on quick wins (high impact, low effort).

- **A/B Test Ideas:** For each key action, design a controlled test. E.g., test whether free shipping upsell increases CLV more than 10% discount; or whether an email nurture sequence improves repeat purchase rate.

- **Image Suggestions:**

- *Team discussion:* ("team planning unsplash", **Unsplash**, embed).
- *Sales graph:* ("sales growth chart unsplash", **Unsplash**).
- *Priority matrix:* ("strategy planning whiteboard unsplash", **Unsplash**).

Filename: `global_superstore_strategies.pdf` (≈3 pages).
RAG Chunking: 450 tokens, 50 overlap; metadata: {`source`:"strategies", `page`, `section`}.

Each PDF is structured for clarity and easy conversion. The images above are suggestions; the actual PDFs should include similar royalty-free visuals with captions. All metrics and strategies above are drawn from industry sources[1][11][8], ensuring analytical rigor.

ZIP Delivery & RAG Ingestion: We will compile the PDFs into a ZIP file with filenames as above. For RAG/embedding, we recommend chunk sizes ~450–500 tokens with ~10% overlap, and include metadata fields (`source`, `section`, `page`) in each chunk for context. Each PDF's content can be split by section or page as needed. After download, ingest the ZIP so the RAG system can retrieve specific sections by metadata.

Sources: Authoritative references used include Tableau community content, Kaggle/Medium dataset descriptions[2][3], and industry KPI guides[8][6][12][1][11]. Each KPI formula and insight is backed by these sources.